

## Standards for LED products

### IES (Illuminating Engineering Society of North America)

- LM-79-08 for electrical and photometric measurements of solidstate lighting products
- LM-80-08 for measuring the lumen maintenance of LED light sources
- RP-16-05 for adding SSL product terms
- TM-16-05 tech memorandum of SSL

### CIE (Commission Internationale de l'Eclairage)

- 127-2007 (for single low-power LEDs only)
- CIE-127 (2007) A&B (for intensity measurement of discrete LEDs)

### ASSIST (RPI Lighting Research Center)

- Recommendation for testing and evaluating luminaires used in directional lighting

### ANSI (American National Standards Institute)

- C78.377-2008 for chromaticity of SSL sources (indoor only)

### DOE (US Department of Energy)

- ENERGY STAR program requirements for SSL luminaires

labs or production lines.

Spatial distribution is a measurement of intensity as a function of viewing angle. This measurement is traditionally performed by holding the test lamp stationary and swinging a small-aperture detector in an arc about the device under test. A complete hemispherical characterisation of the device can be assembled by rotating the plane of measurement with respect to the device, and repeating this test for a number of meridional angular scans.

For mapping the angular characterisation, you need to manually rotate the LED. However, new assemblies have an automated axial range that eliminates the need for manual rotation. New equipment also feature high-speed USB interface to capture and store data points on a computer.

## Life-span measurement

LEDs are predicted to work for 100,000 hours, while LED products last 70,000-80,000 working hours. To actually test it, will you keep the luminaire on for 24 hours a day for eight-ten years? Well, not possible practically. Most of the test results and conclusions are approximated as the function of some other parameter. "Lifetime of LED products is determined as a measurement that is function of temperature," explains Mahajan. "Thermal evaluation or checking optical parameters as a function of

temperature is critical to predict the life of the product."

"As it is practically impossible to perform life-test of an LED for ten years, accelerated tests are carried on the products. Characteristics of the product are monitored for extreme conditions. So if the product is to work at 50°C, we take it to the oven and test it for 150°C. The graph of LED reaction helps in determining the approximate life. Thus the life span of the product is concluded," says Arora.

Agreeing to this, Chui says, "Tests and measurements solutions for time-consuming parameters like lumen maintenance (LM) and lifespan (LS) of LED products eventually conclude with mathematical or statistical predictions or estimations. To end up with more credible T&M results from LM and LS, more elaborate mathematical models or statistical analyses should be invoked to perform more exact and precise predictions or estimations."

While explaining how the tests are actually conducted, Arora says, "If the LED is recommended by the manufacturer for 700mA current, the actual tests are performed at 50 per cent of the rated value, i.e., 350 mA or 450 mA. Practical conditions such as voltage fluctuations and surges contribute to the overall life of the LED. Based on this, if the LED manufacturer talks of 100,000 hours, to be on very conservative side, take 50 per cent of

the rated life."

You can have best design and best luminosity, but if you don't have proper heat management, it will affect the life-span of the product.

## Colour measurement and consistency

"One of the most important parameters of LED products along with the total luminous flux is colour quantities. It is important to understand the colour-dominant wavelength, chromaticity and correlated colour temperature," explains Mahajan. Integrating spheres along with spectroradiometer are used to measure colour quantities.

Colour quantity measurements are most uncertain, and may result in large unexpected results. National Institute of Standards and Technology (NIST) has established a reference spectroradiometer, tailored for LED colour measurement and spectral colour measurement. Spectroradiometers are used to measure spectral power distribution in the visible region of wavelengths. Charge-coupled-device (CCD) spectrometers are sensitive to a wide range of wavelengths, from infrared and optical to ultraviolet and X-rays. Some of the test stations have fast-scanning 16-bit, 2048-element CCD spectrometer configured to cover wavelength in the range of 280 nm to 800 nm.

"Normally, red-green-blue (RGB) and intensity tests are performed on LEDs. The LED analyser allows you to perform fast and automatic tests of both the colour and intensity (brightness) of the LED. In this, the light from LEDs is collected by plastic optical fibres and carried to the LED analyser board for measurement and analysis," shares Jim Hill, director of sales and marketing, Intrinsic Quality.

## Eye safety tests for LEDs

As quoted on the website of Orb Optonix, "A system determined to be safe may fall into a different classification with addition of an optic